

$$\textcircled{1} \quad \underline{\underline{4.3-4}}$$

$$E = \|Ax - b\|^2$$

$$\Rightarrow Ax - b \Rightarrow \begin{bmatrix} A_1x - b \\ A_2x - b \\ \vdots \end{bmatrix}$$

$$\|Ax - b\|^2$$

$$= (A_1x - b_1)^2 + (A_2x - b_2)^2 \dots$$

$$\Rightarrow (A_1x)^2 + (b_1)^2 - 2(A_1x)(b_1) \dots$$

$$\Rightarrow \|Ax\|^2 + \|b\|^2 - 2(A_1x b_1 + A_2x b_2 \dots)$$

$$\|Ax\|^2 + \|b\|^2 - 2(Ax \cdot b) = E$$

$$\boxed{Ax = c}$$

$$c^2 + \|b\|^2 - 2cb = E$$

$$\frac{dE}{dc} = 2c - 2b = 0$$

$$\boxed{c = b}$$

$$E_1 = (A_1x - b_1)^2$$

$$E_2 = (A_2x - b_2)^2$$

:

$$E_4 = (A_4x - b_4)^2 = (C + 4D - 20)^2$$

$$\Rightarrow C^2 + 16D^2 + 400 - 40C + 8CD - 160D = 0$$

$$2C + 8D - 40 = 0$$

$$\boxed{C + 4D = 20}$$

$$C + 4D = 20$$

$\Rightarrow$ A was not given in the question but apparently we just have to assume an A.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} \quad x = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$A_4 x - b_4 \Rightarrow (c + 4d - 20)^2 \quad \boxed{c + 4d = 20}$$

$$A_3 x - b_3 \Rightarrow (c + 3d - 8) = 0$$

$$c + 3d - 8 = 0$$

$$\boxed{c + 3d = 8}$$

$$c + d = 8$$

$$E' = \begin{bmatrix} c \\ c + d - 8 \\ c + 3d - 8 \\ c + 4d - 20 \end{bmatrix}$$

$$c^2 + (c + d - 8)^2 + (c + 3d - 8)^2 + (c + 4d - 20)^2$$

$$\Rightarrow c + 2d = 9$$

$$\Rightarrow 8c + 26d = 112$$

② 4.3-6

$$b = [0 \quad 0 \quad 0 \quad 20]$$

$$a = [1, 1, 1, 1]$$

$$\hat{x} = \frac{a^T b}{a^T a} a$$

$$= 9 a = [9, 9, 9, 9]$$

$$c = [-9, -1, -1, 11]$$

$$\boxed{c^T a = 0}$$

$$\|c\| = \sqrt{81 + 1 + 1 + 121}$$

$$= \sqrt{204} = 2\sqrt{51}$$

$$\textcircled{3} \quad \underline{\underline{4.6 - 7}}$$

$$D = \frac{112}{26} \Rightarrow \frac{56}{13}$$

$$b = Dt$$

$$D \cdot 0 = 0$$

$$D \cdot 1 = 8$$

$$D \cdot 3 = 8$$

$$D \cdot 4 = 20$$

$$A = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$C + Dt \rightarrow Ca_1 + Da_2$$

$$A^T A = 26$$

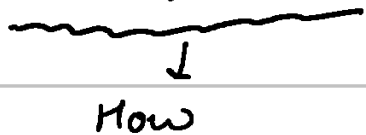
$$A^T b = 112$$

$$D = \frac{56}{13}$$

④ $n \cdot 3 - 9$

$$b = C + Dt + E \cdot t^2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 64 \end{bmatrix}$$



How

$\Rightarrow$ What would be our third column?

We square & add the second row because that is the "t" row

⑥ 4.3-26

$$b = (0, 1, 3, 4)$$

$$(1, 0) \quad (0, 1) \quad (-1, 0) \quad (0, -1)$$

$$\# \quad C + D = 0$$

$$C + E = 1$$

$$C - D = 3$$

$$C - E = 4$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} z = \begin{bmatrix} 8 \\ -3 \\ -3 \end{bmatrix}$$

$$2E = -3$$

$$\hat{E} = -3/2$$

$$\hat{D} = -3/2$$

$$\boxed{\hat{C} = 2}$$

⑥ 4.3-29 ✓

$x \rightarrow 0, a_1, \dots, a_{n-1}$ $\mathbb{R}^n$
n given points

$\Rightarrow$ One $(n-1)$ dimensional subspace

$\Rightarrow$ Unless they are linearly dependent. ✓

$$② \quad \underline{\underline{4 \cdot 4 = 16}}$$

$$a = (4, 5, 2, 2)$$

$$b = (1, 2, 0, 0)$$

$\Rightarrow$ b's projection on a

$$\frac{a^T b}{a^T a} a \rightarrow \frac{14}{16+25+4+4} a$$

$$\rightarrow \frac{14}{49} a$$

$\Rightarrow$ orthonormal vectors in a & b

$$q_1 = \frac{a}{\|a\|}$$

$$B = b - p \Rightarrow$$

$$(1, 2, 0, 0) - \left(\frac{56}{49}, \frac{76}{49}, \frac{28}{49}, \frac{28}{49} \right)$$

$$\Rightarrow \left(\frac{7}{49}, \frac{27}{49}, -\frac{21}{49}, -\frac{28}{49} \right) = B$$

$$⑧) \quad \underline{\underline{4 \cdot 4 - 10}}$$

$$a) \quad [c_1 \quad c_2 \quad c_3] \cdot \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = 0$$

$$q_1^T q_2 = 0$$

$$q_2^T q_3 = 0$$

$$q_3^T q_1 = 0$$

$$\Rightarrow c_1 q_1 \cdot q_1 + 0 + 0 = 0$$

$$\boxed{c_1 = 0}$$

$$b) \quad Qx = 0$$

$$Q^T \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix} x = 0$$

$$\underline{\underline{Ix = 0}}$$

$$(9) \quad 4.4 - 18$$

$$a = (1, -1, 0, 0)$$

$$b = (0, 1, -1, 0)$$

$$c = (0, 0, 1, -1)$$

$$A = a/\|a\|$$

$$B = b - \frac{a^T b}{a^T a} a \Rightarrow \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix}$$

$$C = c - \frac{a^T c}{a^T a} a - \frac{B^T c}{B^T B} B$$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1/3 \\ 1/3 \\ -2/3 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix}$$

⑩ 4.4-35

$$[Q, R] =$$

$$q_1 = [1, -1, 0, 0]$$

$$q_2 = [1, 1, -2, 0]$$

$$q_3 = [1, 1, 1, -3]$$

$$q_4 = [1, 1, 1, 1]$$

$$(ii) \quad \underline{\underline{4 \cdot 4 = 16}}$$

$$A \rightarrow m \times n$$

$$\text{rank } A = n$$

$$\begin{bmatrix} Q_1 & Q_2 \end{bmatrix}_{m \times m} \begin{bmatrix} R \\ 0 \end{bmatrix}_{m \times n}$$

$$Q_1 \rightarrow \text{column space}$$

$Q, R \rightarrow$ just a linear combination of columns

$$A^T \rightarrow \begin{bmatrix} R & 0 \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix}$$

$$Q_2 \perp Q \rightarrow \underline{\underline{N(A^T)}}$$

$$\textcircled{12} \quad \underline{\underline{5.1 - 10}}$$

$$Ax = 0$$

$$A \rightarrow \sum_{i=1}^n a_{ki} = 0 \quad \forall k$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & - & - & \vdots \\ a_{n1} & - & - & a_{nn} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = 0$$

$$= \begin{bmatrix} a_{11} + a_{12} & - & a_{1n} \\ \vdots & & \end{bmatrix} = 0$$

$$x = [1, 1, \dots, 1]$$

$N(A)$ is at least one dimensional.

$$\boxed{\det(A) = 0}$$

(13) 5.1 - 18

$$\det \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & 0 & (c^2-a^2) - \frac{(b^2-a^2)(c-a)}{b-a} \end{bmatrix}$$

$$\Rightarrow 1 \cdot (c^2-a^2 - (b^2-a^2) \frac{c-a}{b-a})$$

$$\Rightarrow (c-a) \left((c+a) - (b^2-a^2) \frac{1}{b-a} \right)$$

$$(c-a) (b-a) (c+b-a) - (c-a) (b-a) (b-a)$$

$$= (c-a)(c-b)(b-a)$$

(14) $\underline{\underline{5 - 1 - 51}}$

pivot of $\text{hilb}(5)$

$\Rightarrow 1, \frac{1}{12}, -\frac{1}{180}, \frac{1}{1400}, -\frac{1}{83200}$

⑬ $5 \cdot 1 - 32$

$\Rightarrow \text{rand}(n) \rightarrow \text{calculable}$

$\text{rand}_n(n) \rightarrow \text{Inf for bigger } n;$